

Hao-Chun Liang

[Email](#) | [Linkedin](#) | 0982-955885

EDUCATION

National Yang Ming Chiao Tung University

M.Eng. in IPSI (ICS & EDA Group) | GPA: 4.3/4.3

Sep. 2025 – Jun. 2027 (Exp.)

Hsinchu, Taiwan

National Tsing Hua University

B.S. in Electrical Engineering | GPA: 4.1/4.3 | Distinction Graduation

Sep. 2020 – Jun. 2024

Hsinchu, Taiwan

- Transferred from Dept. of Physics (Top 1 of 65; College of Science Elite Student Award).

Graduate Coursework (NYCU, all A+): IC Design Laboratory (ICLAB), DSP VLSI Design, Deep Learning, Stochastic Process

Selected Undergraduate Coursework (NTHU): IC Design Lab (A+), VLSI Testing, VLSI Design, Computer Architecture, Digital Communication System (A+)

WORK EXPERIENCE

Digital IC Design Intern (Returning)

Andes Technology Corporation

Jul. 2026 (Exp.)

Hsinchu, Taiwan

Graduate Research Assistant

National Yang Ming Chiao Tung University | Parallel Computing System Laboratory

Sep. 2025 – Jun. 2027 (Exp.)

Hsinchu, Taiwan

Digital IC Design Intern

Andes Technology Corporation | Acceleration R&D Division

Nov. 2024 – Aug. 2025

Hsinchu, Taiwan

AWARDS & CERTIFICATES

Elite New Graduate Scholarship

Industry Academia Innovation School, NYCU

Dec. 2025

Hsinchu, Taiwan

Excellent Poster Award

Taiwan and Japan Conference on Circuits and Systems (TJCAS)

Aug. 2025

Kaohsiung, Taiwan

Cell-Based Digital Chip Design and Implementation (ADFP - TSMC 16nm)

Taiwan Semiconductor Research Institute (TSRI)

Sep. 2024

Hsinchu, Taiwan

Distinction Graduation (Top 5% of 116 in Professional Elective GPA)

National Tsing Hua University | Electrical Engineering

Jun. 2024

Hsinchu, Taiwan

College of Science Elite Student Award (Top 1 of 65)

National Tsing Hua University | College of Science

Apr. 2021

Hsinchu, Taiwan

Presidential Award (Received 3 times)

National Tsing Hua University

Sep. 2020 – Jun. 2024

Hsinchu, Taiwan

PUBLICATIONS

UniTrac: A Unified FPGA Accelerator for GNN-based Particle Track Reconstruction in HEPs

ACM Transactions on Reconfigurable Technology and Systems (In-Submission) | Second + Corresponding Author

Real-Time GPU Kalman-Filter Tracking via Kernel Refactoring and INT8 Surrogates

Fast Machine Learning for Science Conference 2025, CERN | First Author

An Integrated FPGA Implementation of Complete GNN-Based Trajectory Reconstruction

Taiwan and Japan Conference on Circuits and Systems (TJCAS) 2025 | Second Author

A High-Performance Implementation of GNN-Based Trajectory Reconstruction on FPGA

VLSI Design/CAD Symposium 2025 | Second Author

Fine-Grained Image Recognition from Scratch with Teacher-Guided Data Augmentation

arXiv:2507.12157 | Fifth Author

PROJECTS

Regfile Automatic Code Generation

Andes Technology Corporation

Nov. 2024 – Aug. 2025

Hsinchu, Taiwan

UltraLightweight HDC-CNN for NeurIPS 2025 Weak Lensing Challenge

NeurIPS FAIR Universe Challenge

Oct. 2025 – Nov. 2025

Top-20 Ranking

NVIDIA NSYS/NCU Profile and Analysis of Token Reduction for ViT

National Yang Ming Chiao Tung University | Parallel Computing System Laboratory

Oct. 2025 – Nov. 2025

Hsinchu, Taiwan

RISC-V 5-Stage Pipelined CPU Core

National Tsing Hua University

Dec. 2023 – Jan. 2024

Hsinchu, Taiwan

SKILLS

Programming Languages: Python, C/C++, Verilog/SystemVerilog, HLS, CUDA, Perl, Shell

Domains: Efficient Machine Learning, Computer Architecture, Digital IC Design, RTL/C++ Codegen